KUBERNETES-APPLICATION

Step -1: Before writing any yaml files for application. You must clone the GitHub file by using command

git clone https://github.com/Dudduvenkatesh/fss-Retail-App_kubernetes.git . This command clones the all files and dependencies and libraries.

```
controlplane:~$ git clone https://github.com/Dudduvenkatesh/fss-Retail-App_kubernetes.git
Cloning into 'fss-Retail-App_kubernetes'...
remote: Enumerating objects: 2424, done.
remote: Counting objects: 100% (134/134), done.
remote: Compressing objects: 100% (96/96), done.
remote: Total 2424 (delta 77), reused 42 (delta 34), pack-reused 2290 (from 2)
Receiving objects: 100% (2424/2424), 9.40 MiB | 9.41 MiB/s, done.
Resolving deltas: 100% (549/549), done.
controlplane:~$
```

Step -2: Click on the file directory by using command

cd **fss-Retail-App_kubernetes**/ this command redirects to this location. Then click on **ls** command to see the all files and dependencies and libraries.

```
controlplane:~$ ls
filesystem  fss-Retail-App_kubernetes
controlplane:~$ cd fss-Retail-App_kubernetes/
controlplane:~/fss-Retail-App_kubernetes$ ls
Dockerfile  Public     docker-compose.yaml  node_modules       package.json  test-volumes.yaml
Notes       README.md  k8s-manifests        package-lock.json  server.js
controlplane:~/fss-Retail-App_kubernetes$ cd k8s-manifests/
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ ls
mongodb-deployment.yml  mongodb-service.yml  usernode-js-service.yml  userprofile-deployment.yml
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$
```

Step – 3: Create a namespace using command

kubectl create ns namespace – this command creates the namespace.

Note: If we can't create the namespace then all the yaml files get created in by default namespace.

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl create ns venkatesh-ns
namespace/venkatesh-ns created
```

Step -4 : Change my filename to understanding purpose

mv oldfilename newfilename – this command change the filename

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ ls
mongodb-deployment.yml  mongodb-service.yml  usernode-js-service.yml  userprofile-deployment.yml
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ vi configmap.yaml
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl create ns venkatesh-ns
namespace/venkatesh-ns created
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ mv mongodb-deployment.yml retail-mongodb-deployment.yml
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ mv mongodb-service.yml retail-mongodb-service.yml
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ mv usernode-js-service.yml retail-app-deployment.yml
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ mv userprofile-deployment.yml retail-app-service.yml
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ ls
configmap.yaml           retail-app-service.yml         retail-mongodb-service.yml
retail-app-deployment.yml  retail-mongodb-deployment.yml
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$
```

Step -5: Create config.yaml file

Configmap means it used to store the non-sensitive data in the form of key-value pairs.

Without ConfigMap:

- Hardcode values in the Deployment file

  Or hardcode values inside the application code

With ConfigMap:

- Just update configuration
  Restart deployment
  No image rebuild needed

Configmap.yaml

apiVersion: v1

kind: ConfigMap

metadata:

  name: retail-app-config

  name: venkatesh-ns

data:

  MONGODB_URI: “mongodb://localhost:27017/myDatabase”

  SESSION_SECRET: “1234”

  PORT: “3130”

  MONGO_INITDB_DATABASE: “myDatabase”

To run this yaml file use this command – kubectl apply -f <yaml-filename>

To check this yaml file created or not by using this command – kubectl get <yaml-filename> -n <namespace>

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl apply -f configmap.yaml
configmap/retail-app-config created
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl get configmap -n venkatesh-ns
NAME                DATA   AGE
kube-root-ca.crt    1      8m48s
retail-app-config   4      7s
```

Step-6 : Create a secret-yaml file

In **Kubernetes**, a **Secret** is an object used to store **sensitive information** securely.

Sensitive information converts into encoded value by using base64 by using command

echo -n 'filename' | base64 - for encoding the data

echo -n 'filename' | base64 –decode – for decoding the data.

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ echo -n 'chagantyteja2502@gmail.com' | base64
Y2hhZ2FudHl0ZWphMjUwMkBnbWFpbC5jb20=
```

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ echo -n 'yxoq bjuk rdnt alzp' | base64
eXhvcSBianVrIHJkbnQgYWx6cA==
```

Secret.yaml:

apiVersion: v1

kind: Secret

metadata:

  name: retail-app-secret

  namespace: venkatesh-ns

type: Opaque

data:

  EMAIL_USER: Y2hhZ2FudHl0ZWphMjUwMkBnbWFpbC5jb20=

  EMAIL_PASS: eXhvcSBianVrIHJkbnQgYWx6cA==

Command: Kubectl apply -f <filename>

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl apply -f secret.yaml
secret/retail-app-secret created
```

Command : kubectl get <filename> -n <namespace>

```
controlplane:~/fss-Retail-App kubernetes/k8s-manifests$ kubectl get secret -n venkatesh-ns
NAME                TYPE     DATA   AGE
retail-app-secret   Opaque   2      29s
```

Step 7: Create a mongodb deployment file

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: retail-mongodb
  namespace: venkatesh-ns
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mongodb
  template:
    metadata:
      labels:
        app: mongodb
    spec:
      containers:
      - name: mongodb
        image: mongo:latest
        ports:
        - containerPort: 27017
        env:
        - name: MONGODB_URI
          valueFrom:
            configMapKeyRef:
              name: retail-app-config
```

key: MONGODB_URI

Command: Kubectl apply -f <filename>

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl apply -f retail-mongodb-deployment.yml
deployment.apps/retail-mongodb created
```

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl get all -n venkatesh-ns
NAME                                 READY   STATUS    RESTARTS   AGE
pod/retail-mongodb-76d69799bc-92v8d  1/1     Running   0          97s

NAME                            READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/retail-mongodb  1/1     1            1           97s

NAME                                       DESIRED   CURRENT   READY   AGE
replicaset.apps/retail-mongodb-76d69799bc  1         1         1       97s
```

Step 8: Create retail-mongo-service.yaml

apiVersion: v1

kind: Service

metadata:

  name: mongodb-service

  namespace: venkatesh-ns

spec:

  selector:

    app: mongodb

  ports:

    - port: 27017

      targetPort: 27017

      protocol: TCP

  type: ClusterIP

Command: Kubectl apply -f <filename>

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl apply -f retail-mongodb-service.yml
service/mongodb-service created
```

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl get all -n venkatesh-ns
NAME                                READY   STATUS    RESTARTS   AGE
pod/retail-mongodb-76d69799bc-92v8d   1/1     Running   0          2m43s

NAME                      TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)     AGE
service/mongodb-service   ClusterIP   10.103.34.166   <none>        27017/TCP   16s

NAME                             READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/retail-mongodb   1/1     1            1           2m43s

NAME                                        DESIRED   CURRENT   READY   AGE
replicaset.apps/retail-mongodb-76d69799bc   1         1         1       2m43s
```

Step -9: Create retail-app-deployment and retail-app-service

Retail-app-deployment.yaml:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: retail-deployment
  namespace: venkatesh-ns
spec:
  replicas: 4
  selector:
    matchLabels:
      app: retailapp
  template:
    metadata:
      labels:
        app: retailapp
    spec:
      containers:
      - name: retail-container
        image: saiteja2502/userprofileretail:latest
        ports:
        - containerPort: 3130  # Change to match your app's port
        env:
```

```yaml
        - name: MONGODB_URI
          valueFrom:
            configMapKeyRef:
              name: retail-app-config
              key: MONGODB_URI
        - name: SESSION_SECRET
          valueFrom:
            configMapKeyRef:
              name: retail-app-config
              key: SESSION_SECRET
        - name: PORT
          valueFrom:
            configMapKeyRef:
              name: retail-app-config
              key: PORT
        - name: EMAIL_USER
          valueFrom:
            secretKeyRef:
              name: retail-app-secret
              key: EMAIL_USER
        - name: EMAIL_PASS
          valueFrom:
            secretKeyRef:
              name: retail-app-secret
              key: EMAIL_PASS
```

retail-app-service.yaml:

```yaml
apiVersion: v1
kind: Service
metadata:
```

```yaml
  name: retail-service
  namespace: venkatesh-ns
spec:
  type: ClusterIP  # Use NodePort if LoadBalancer is not supported
  ports:
    - port: 3130
      targetPort: 3130
      protocol: TCP
  selector:
    app: retailapp
```

o/p for retail-app-deployment:

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl apply -f retail-app-deployment.yml
deployment.apps/retail-deployment created
```

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl get all -n venkatesh-ns
NAME                                     READY   STATUS              RESTARTS   AGE
pod/retail-deployment-84886b8898-5jw5f   0/1     ContainerCreating   0          16s
pod/retail-deployment-84886b8898-7lsq6   0/1     ContainerCreating   0          16s
pod/retail-deployment-84886b8898-lpdb7   0/1     ContainerCreating   0          16s
pod/retail-deployment-84886b8898-nnhf9   0/1     ContainerCreating   0          16s
pod/retail-mongodb-76d69799bc-92v8d      1/1     Running             0          3m29s

NAME                      TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)     AGE
service/mongodb-service   ClusterIP   10.103.34.166   <none>        27017/TCP   62s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/retail-deployment   0/4     4            0           16s
deployment.apps/retail-mongodb      1/1     1            1           3m29s

NAME                                           DESIRED   CURRENT   READY   AGE
replicaset.apps/retail-deployment-84886b8898   4         4         0       16s
replicaset.apps/retail-mongodb-76d69799bc      1         1         1       3m29s
```

o/p for retail-app-service:

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl apply -f retail-app-service.yml
service/retail-service created
```

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl get all -n venkatesh-ns
NAME                                     READY   STATUS    RESTARTS      AGE
pod/retail-deployment-84886b8898-5jw5f   0/1     Error     0             83s
pod/retail-deployment-84886b8898-7lsq6   0/1     Error     0             83s
pod/retail-deployment-84886b8898-lpdb7   1/1     Running   1 (18s ago)   83s
pod/retail-deployment-84886b8898-nnhf9   1/1     Running   1 (19s ago)   83s
pod/retail-mongodb-76d69799bc-92v8d      1/1     Running   0             4m36s

NAME                         TYPE           CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
service/mongodb-service      ClusterIP      10.103.34.166   <none>        27017/TCP        2m9s
service/retail-service       LoadBalancer   10.98.120.24    <pending>     3130:32657/TCP   18s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/retail-deployment   2/4     4            2           83s
deployment.apps/retail-mongodb      1/1     1            1           4m36s

NAME                                           DESIRED   CURRENT   READY   AGE
replicaset.apps/retail-deployment-84886b8898   4         4         2       83s
replicaset.apps/retail-mongodb-76d69799bc      1         1         1       4m36s
```

Step 10: The some of the pods showing error in that we need to check the logs and describe.

For this we use 2 commands:

Kubectl logs <podname> -n <namespace>

Kubectl describe pod <pod-name> -n < namespace>

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl describe pod retail-deployment-84886b8898-5jw5f -n ven
katesh-ns
Name:             retail-deployment-84886b8898-5jw5f
Namespace:        venkatesh-ns
Priority:         0
Service Account:  default
Node:             controlplane/172.30.1.2
Start Time:       Sat, 21 Feb 2026 16:03:20 +0000
Labels:           app=retailapp
                  pod-template-hash=84886b8898
Annotations:      cni.projectcalico.org/containerID: a747f8f840c79e5f4c85f1f244b005539aa594b2c61c94b528486f614bc67a2e
                  cni.projectcalico.org/podIP: 192.168.0.5/32
                  cni.projectcalico.org/podIPs: 192.168.0.5/32
Status:           Running
IP:               192.168.0.5
IPs:
  IP:           192.168.0.5
Controlled By:  ReplicaSet/retail-deployment-84886b8898
Containers:
  retail-container:
    Container ID:   containerd://0827c4baffdba3e8d1fb1c6b7b86a8e7a7132017d7ee4b272cc2e6f7a92a0bae
    Image:          saiteja2502/userprofileretail:latest
    Image ID:       docker.io/saiteja2502/userprofileretail@sha256:3fbd85b1e1ca8adf4640d342d0ea6de9af85841aca82fa0ba80
e0a411b229001
```

After I restart my app-deployment file using this command

kubectl rollout restart deployment deployment-name -n namespace

Finally to check my pod status and service and deployment and replica set status using this command – kubectl get all -n <namespace>

```
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$ kubectl get all -n venkatesh-ns
NAME                                      READY   STATUS    RESTARTS   AGE
pod/retail-deployment-78545c5c6f-fvnvl    1/1     Running   0          3s
pod/retail-deployment-78545c5c6f-hpngq    1/1     Running   0          5s
pod/retail-deployment-78545c5c6f-q7qcw    1/1     Running   0          3s
pod/retail-deployment-78545c5c6f-rwmtl    1/1     Running   0          5s
pod/retail-mongodb-76d69799bc-92v8d       1/1     Running   0          23m

NAME                        TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)     AGE
service/mongodb-service     ClusterIP   10.103.34.166   <none>        27017/TCP   21m
service/retail-service      ClusterIP   10.98.120.24    <none>        3130/TCP    19m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/retail-deployment   4/4     4            4           20m
deployment.apps/retail-mongodb      1/1     1            1           23m

NAME                                           DESIRED   CURRENT   READY   AGE
replicaset.apps/retail-deployment-78545c5c6f   4         4         4       5s
replicaset.apps/retail-deployment-84886b8898   0         0         0       20m
replicaset.apps/retail-mongodb-76d69799bc      1         1         1       23m
controlplane:~/fss-Retail-App_kubernetes/k8s-manifests$
```